

Analysis and Implementation of a Real-Time Demonstrator on the RTIC Runtime

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Abstract

This paper presents the implementation and analysis of a real-time demonstrator originally developed in *Ada* using the *Ravenscar Profile*, re-implemented in *Rust* using the *RTIC* framework. The demonstrator is designed to run on an *STM32F4* microcontroller and uses a *Fixed-Priority Scheduling* policy combined with variants of *Priority Ceiling Protocol (PCR)* for resource sharing. The analysis is performed using the *MAST* tool to verify timing properties and analyze system performance under various workloads. The results demonstrate the feasibility of adopting *Rust* and *RTIC* for real-time applications, while also highlighting differences and considerations compared to the original *Ada* implementation.

CCS Concepts

• **Software and its engineering** → **Real-time systems software**; **Embedded software**; *Software architectures*.

Keywords

Real-Time Kernels, Real-Time Systems, Embedded Software, Software Architectures

ACM Reference Format:

Gianluca Bresolin and Tommaso Prandin. 2025. Analysis and Implementation of a Real-Time Demonstrator on the RTIC Runtime. In *Proceedings of Real-Time Kernels and Systems Exam (RTKS '25)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Real-time embedded systems demand precise control over timing, concurrency and resource management. Such systems are often used in critical applications where failure to meet timing constraints can lead to severe consequences. Hence, developing a real-time system requires careful consideration of the underlying runtime environment, necessitating a clear understanding of how the system will behave in order to provide sufficient guarantees on meeting timing requirements.

The choice of the runtime environment is a crucial aspect in the design of a real-time system, as it can significantly impact the system's ability to meet its desired behaviour through runtime mechanisms.

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RTKS '25, Padova, Italy

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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/XXXXXXX.XXXXXXX>

One of the most popular languages that meets the needs of real-time systems is *Ada*, which has been widely used in the development of safety-critical systems due to its deterministic constructs that allow high analysability of the code. While *Ada* remains a robust choice, recent advances in the *Rust* programming language have introduced new opportunities for real-time systems development. In this paper, a real-time system originally implemented using the *Ada Ravenscar Profile* is re-implemented in *Rust* using the *Real-Time Interrupt-driven Concurrency (RTIC)* framework [5], which is responsible for triggering the scheduling of tasks through a *Fixed-Priority Scheduling* and the resource sharing in the system under *Stack Resource Policy (SRP)*.

For the analysis of the system, the *MAST* tool [2] is used to verify the timing properties and to analyze the system under different workloads, with the system running on a *STM32F4* microcontroller. The goal of this work is to demonstrate the feasibility of using *Rust* and *RTIC* for real-time systems, while also providing some considerations and comparison with the original *Ada* implementation. The rest of the paper is organized as follows. Section 2 presents a formal description of the system model. Section 3 introduces the *Rust runtime*, in particular the *RTIC* framework and its key concepts. Sections 4 and 5 discuss the measurement of execution times and the detection of deadline misses, respectively. Section 6 provides an overview of the *MAST* tool and its analysis capabilities. In Section 7, we present the *Fixed-Priority Scheduling (FPS)* analysis of the system. Finally, Section 8 concludes the paper reporting the main findings.

2 System Model

The software architecture of the system under test follows the one proposed in "*Guide for the Use of Ada Ravenscar Profile in High Integrity Systems*" [15]. It is comprised by one periodic producer task that activates two other sporadic tasks, depending on certain criteria. Additionally, there is a sporadic event server that manages the occurrence of external events. In the implemented application, events are generated by a periodic task that emulates user interaction, for the sake of repeatability.

Table [1] enumerates the tasks constituting the application, together with their type, deadline, period (if applicable) and priority. Priorities have been assigned using the *Deadline Monotonic* algorithm, which assigns priorities in decreasing order, starting with the task having the smallest relative deadline. *Deadline Monotonic* is an optimal policy for constrained deadlines and fixed-priority scheduling of mutually independent tasks [19, p. 119]. The deadline watchdogs, only implemented in the *Rust* system to cover the missing support from the runtime, have been given the

Task	Task Type	Period / Min. Inter-arrival (ms)	Deadline (ms)	Priority
Activation Log Reader	Sporadic	3000	1000	3
On Call Producer	Sporadic	5000	800	5
Regular Producer	Periodic	1000	500	7
External Event Server	Sporadic (interrupt)	5000	100	11
Interrupt Generator	Periodic	5000	50	13
ALR Deadline Watchdog	Sporadic	3000	1000	15
OCP Deadline Watchdog	Sporadic	5000	800	15
RP Deadline Watchdog	Periodic	1000	500	15
EES Deadline Watchdog	Sporadic (interrupt)	5000	100	15

Table 1: List of tasks in the system, with their parameters

Resource	Users	Ceiling π_R
Activation Log	Activation Log Reader, External Event Server	11
Event Queue	External Event Server, External Event ISR	11
Request Buffer	Regular Producer, On-Call Producer	7
ALR Deadline Object	Activation Log Reader, ALR Deadline Watchdog	15
OCP Deadline Object	On-Call Producer, OCP Deadline Watchdog	15
RP Deadline Object	Regular Producer, RP Deadline Watchdog	15
EES Deadline Object	External Event Server, EES Deadline Watchdog	15

Table 2: List of shared resources in the system, with their users and priority ceilings

highest priority in order to detect overrun without interference from other tasks (see section 5 for further details).

The tasks in the system share various resources. Access to them is regulated by variants of the *Priority Ceiling Protocol* [23] depending on the implementation: the *Ada* system implements the *Immediate Priority Ceiling Protocol* [16], while the *Rust* system uses the *Stack Resource Policy* [12].

While the implementation is slightly different, both protocols show the same properties and response-time behavior. They require to specify the ceiling value for each shared resource, which is obtained by picking the maximum priority value among all the task using the given resource.

Table [2] enumerates all the shared resources in the system, together with their users (tasks) and their priority ceiling.

In order to emulate a realistic workload for task requiring some kind of computation, the small variant of the Whetstone benchmark [17] is used. It includes different kind of computations: integer, floating-point, trigonometry, branches and function calls, with the aim of testing the system under a wide range of loads.

The Whetstone benchmark provides a configurable workload parameter that directly controls the number of iterations performed by the procedure. Each iteration executes a standardized mix of computational operations (floating-point arithmetic, trigonometric functions, loops, and conditional statements), and the benchmark's performance is measured in *Whetstone Instructions Per Second* (WIPS). A *Whetstone instruction* represents a normalized unit of computational work as defined by the original benchmark [17], used to abstract the specific hardware implementation and provide a machine-independent measure of computational load that is representative of typical scientific workloads. One kilo-whetstone

corresponds to 1000 such normalized computational operations. By scaling this parameter, we can adjust the worst-case execution time (WCET) of each task, creating different system utilization levels. For instance, doubling the workload parameter approximately doubles the execution time, allowing us to systematically stress-test the system. This parametric approach enables controlled experimentation where we can observe how the system behavior changes as computational load increases, first by simulating it using the MAST tool [2], then by running the application in a real setting.

To this end, we have used the *Olimex STM32H405* [20] development board, together with a *STLinkv2* [7] debugging probe.

2.1 Tasks Description

2.1.1 Activation Log Reader. The *Activation Log Reader* is a sporadic task which awaits on a signal received from the *Regular Producer*, and then performs its internal operation which consists of a synthetic Whetstone workload, followed by a reading of the status of the *Activation Log* and a print that notifies the end of its sporadic activation.

2.1.2 On-Call Producer. The *On-Call Producer* is a sporadic task that waits for a request deposit into the *Request Buffer*, then reads it and runs a Whetstone workload with load equal to the extracted value. The requests are sent by the *Regular Producer*, and they are meant to represent a scenario in which a main task delegates some work to a helper task in the event of an overload.

2.1.3 Regular Producer. The *Regular Producer* is a periodic task that runs a Whetstone workload with a predefined load value, then, based on a predetermined criterion, can release the *Activation Log Reader* and send a work request to the *On-Call Producer*.

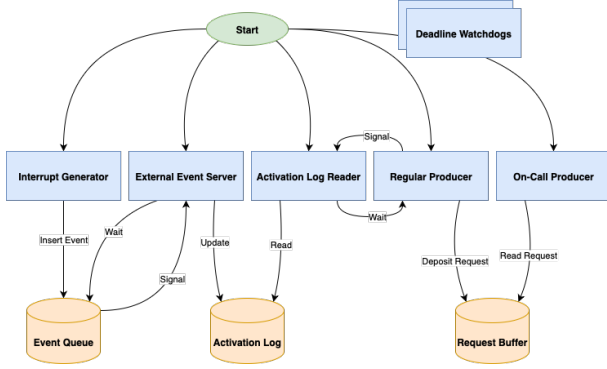


Figure 1: System architecture diagram

2.1.4 External Event Server. The *External Event Server* is a sporadic task that waits for events put onto the *Event Queue* by the interrupt handler bound to the simulated external event. It updates the *Activation Log* counter on each activation.

2.1.5 Deadline Watchdogs. In the Rust system there is no support in the runtime for the detection of tasks' deadline misses. For this reason, we had to manually associate each task in the system with a watchdog monitor task responsible for checking deadline overruns. It has the maximum priority to prevent interference. It maintains the number of deadline misses and whenever it detects one it prints a warning message (see section 5 for further details).

Figure [1] shows a high-level view of the system architecture

3 Rust Runtime

Starting from the demo application provided by the *Ada* guide [15], we implemented an equivalent system fully written in *Rust*.

Unlike *Ada*, which provides a reference runtime implementation, *Rust* delegates this responsibility to third-party developers. This approach offers several advantages: it promotes flexibility and ecosystem development by encouraging competitive runtime development and enabling implementations tailored to specific use cases. However, this approach also causes ecosystem fragmentation and could also result in lower "quality" of the runtimes, considering that a single reference implementation will naturally tend to concentrate resources towards its development from all the actors interested in its usage or analysis, instead of dispersing them across multiple projects.

Thus, when designing and implementing a real-time system in *Rust*, some effort must be spent in order to identify and integrate the most appropriate runtime components for the system, paying attention to select compatible components that can work in harmony together i.e. with compatible APIs and functioning in the chosen runtime environment. In our case this means choosing a *concurrency runtime* that provides the task abstraction and scheduling primitives, a *logging framework* and a *Hardware Abstraction Layer* that provides higher-level abstractions over the system hardware.

3.1 RTIC

As for the *concurrency runtime*, we have decided to use the *RTIC* framework [8]. It is generally considered to be a "concurrency framework", since it does not contain a full-fledged software kernel and relies on external *Hardware Abstraction Layer* to access peripherals.

RTIC implements the *Stack Resource Policy* for shared resource access management [12], with *Fixed-Priority Scheduling* [19, Chapter 6].

SRP has some really useful runtime properties for real-time systems, both for performance and especially for amenability to response-time analysis. Some of those properties are:

- Bounded priority inversion blockage by the longest critical section of a lower priority job;
- Absence of circular-wait deadlocks;
- Tasks may utilize a single shared stack;
- Tasks are guaranteed to have wait-free access to all the resources they need after release;

Those come with the following set of requirements:

- Tasks are associated with static *pre-emption levels*;
- Tasks execute on a single core;
- No non-preemptable resources are held across blocking or yield points;
- Resource must be claimed and released in LIFO order.

The motives behind these requirements and the proof of the properties are outside the scope of this work; they can be found in the original paper [12].

SRP based scheduling requires the set of static priority tasks and their access to shared resources to be known in order to compute a static ceiling π for each resource. The static resource ceiling $\pi(r)$ reflects the maximum static priority of any task that accesses the resource r . This is the reason why, despite its excellent properties, is not used in general-purpose OS: it requires the resource dependency graph to be known statically, which is generally not possible. In addition to this, the protocol has a runtime cost associated to every resource access and release, that involves updating the current system ceiling $\pi_s(t)$; and also at every job release to compare its priority against the system ceiling.

FPS is a very simple scheduling policy: every task is statically given a fixed priority value (values must have an order), and all jobs released will inherit the priority of their task. At every point in time the running job is the one with the highest priority among all the *ready* ones.

The key idea of *RTIC* is to use the native features provided by the interrupt engine of ARM microcontrollers (*NVIC* [11]) to implement these two protocols, obtaining a *hardware-accelerated runtime* suitable for real-time systems, with minimal overhead.

In the case of the ARM Cortex-M architecture, each interrupt vector entry v_i is associated an *Interrupt Service Routine* isr_i , a static priority (p_i), an enabled- (en_i) and a pending-bit ($pend_i$).

An interrupt i is scheduled (run) by the hardware only if both of the following conditions are met:

- (1) i is pending and enabled and has a priority higher than the BASEPRI register;
- (2) i has the highest priority among interrupts meeting 1.

The *SRP* model for single-core static scheduling, on the other hand, states that a task should be scheduled (run) under the conditions:

- (1) It is requested to run and has a static priority higher than the current system ceiling (Π);
- (2) It has the highest static priority among tasks meeting 1.

The system ceiling Π is computed as the maximum priority ceiling of the currently held resources.

This shows a very close relationship between the scheduling logic implemented in hardware by the NVIC, and the model required by *SRP*. The mapping between logical tasks and actual hardware is quite straightforward:

- Each task t is mapped to an interrupt vector index i with the corresponding ISR $isr_i = t$ and given the correct static priority $p_i = p(t)$;
- The system priority ceiling Π is mapped to the BASEPRI register, which is updated at the entry and exit of every *critical section* (i.e. lock).

The mapping between tasks and ISR, and resources to their static ceilings is computed at compile-time by *RTIC* using *procedural macros*, thus it has no impact on the performance at runtime.

Actually, *RTIC* maps each task priority level to an ISR at the corresponding priority. Each ISR runs an *asynchronous executor* that handles tasks wrapped as *async* functions assigned to that priority level. This allows for *async/await* style programming in tasks, lifting the run-to-completion requirement for them, since each section of code between two yield points can be seen as an individual task. The compiler enforces that no *await* operations can occur while holding a resource, preventing violations of the strict LIFO requirement on resource usage mandated by *SRP* [8]. However, this compile-time guarantee only applies to statically analyzed code; externally compiled procedures can bypass this verification. The support for this style of programming simplifies the development of complex tasks by automatically managing state transitions between suspension points, eliminating the need for manual state machine implementations when handling multi-stage operations, such as inter-task communication and hardware interactions.

3.2 Defmt

The second piece of the system is *defmt* [1]: it is a logging framework for embedded applications, written in *Rust*.

It uses a novel approach to improve space and computational efficiency of the logging on the embedded device side. Instead of storing the format strings and perform parameter formatting on the device, it *defers* those operations, moving them on the connected host.

Technically, it embeds metadata about format string and the parameters types on a linker section that gets stripped before placing the firmware into the device flash. Thus, the device only needs to send the format string index and raw parameters; then the connected host, provided with the non-stripped binary file, will perform formatting and printing.

The disadvantage of this approach is that without the metadata embedded in the binary file, the messages from the device are not interpretable. Nonetheless, the advantages in terms of size

and computational requirements outweighs the downsides in most cases.

Defmt supports different *backends* for communication between the host and the device, such as *RTT*, *semihosting* and *ITM*; in our implementation we opted to use *RTT*, as we had problems with the *semihosting* backend.

3.3 HAL

The final key component of the system is the *Hardware Abstraction Layer (HAL)*. It provides a high-level, portable interface to interact with the underlying hardware peripherals, abstracting away the low-level register manipulation and platform-specific details.

In the *Rust* embedded ecosystem, *HALs* are typically implemented as crates that follow the *embedded-hal* trait specifications [9]. This standardization allows for interoperability between different hardware platforms and peripheral drivers, enabling code reuse and ecosystem compatibility.

For our implementation, we used the *STM32F4xx HAL* crate [10], which provides abstractions for *GPIO* pins, timers, *UART* communication, *SPI*, *I2C*, and other peripherals specific to the *STM32F4* microcontroller family. The *HAL* ensures type-safe access to hardware resources at compile time, preventing common errors such as accessing uninitialized peripherals or conflicting resource usage, enforced by *Rust* strong type system. In general *RTIC* is *HAL* agnostic: it is possible to use other implementations from different vendors (e.g. *Embassy*).

The most important peripheral needed for the experiment is the timer, which is used to provide the notion of time to the system, via a global *rtic-monotonics* object, thus enabling time-based operations (e.g. delays, periodic tasks, deadline monitoring). *RTIC* has built-in support for our chosen *HAL*, hence implementing the required timer object only required calling a simple macro [6].

4 Execution Times

4.1 Rust

Obtaining the *Worst-Case Execution Times* for every operation identified in the analysis of the system is a crucial step in order to perform response-time analysis.

We decided to leverage the *Data Watchpoint and Trace Unit (DWT)* [11, Chapter 9]. *DWT* supports multiple functionalities such as: debugger hardware watchpoints, multiple comparators, and cycle counters. For our purposes, the latter was sufficient.

In particular, the *DWT* peripheral holds a cycle counter that can be read from the *DWT_CYCCNT* register, which in turn can be used to implement a simple *stopwatch* that can measure time spans.

The chosen *HAL* [10] already provides a stopwatch implementation that correctly handles the wrap around of the clock counter. The typical system clock frequency of our board is 168 MHz, which makes the period $T = \frac{1}{f} \approx 5.95$ ns. Since the counter size is 32 bits, it will wrap around every $2^{32} \cdot T \approx 25$ s, thus it must be handled to prevent wrong readings.

The stopwatch implementation provides an API that closely resembles the workings of a manual stopwatch: it exports a *reset* function that simply clears every stored time, and a *lap* method that stores the current cycle count into a buffer passed during


```

pub trait Profiler {
    fn reset(&mut self);
    fn lap(&mut self);
    fn lap_time(&self, lap: usize) ->
        Option<ProfilingDuration>;
    fn update_wcet(&mut self);
    fn log(&self);
}

```

Listing 1: Profiler trait used to extract WCET

initialization. Lap times can be simply obtained by subtracting successive cycle counts, having care of handling wrap around.

Since this implementation is really basic, we have designed a higher-level abstraction that allows us to obtain the WCET of selected code sections, corresponding to the operations under analysis. Listing [1] shows the *trait* definition for our profiler abstraction. The *lap* method gets called at the end of every iteration (i.e. job execution), using the saved laps; then results can be logged using the corresponding method.

Every different task gets a tailored implementation of the profiler *trait*; they mainly differ on the implementation of the *update_wcet* method, to accommodate the different operations inside the tasks.

The profiling instrumentation is gated behind cargo *features*: when the user desires to start the profiling of a certain operation it needs to enable the related feature at compile-time. This ensures that the application is completely devoid of profiling objects and instructions that are not needed during normal operation, improving performance and code size.

The runtime does not provide a *task time* like Ada Ravenscar, hence, to obtain the desired WCET, we had to manually raise the priority of the task under analysis in order to prevent any interference that could invalidate our time measurements. This is not a very clean solution, but implementing a proper task execution time would require non-trivial modifications to the RTIC runtime, which were considered out of the scope of the work and too time-consuming. Nonetheless, the WCET obtained remain valid.

We have decided to take the maximum execution time among one-thousand executions; when the correct profiling feature is enabled, the tasks are configured to automatically log the WCET results accumulated by the profiler and then panic to stop execution.

4.2 Ada

For the Ada system, we have re-utilized the instrumentation implemented by J. Hu and A. Gobbo in their work [18], aimed to perform response-time analysis over the *Ada* system model, as shown in the original paper [15].

4.3 WCET Times

This section contains the obtained WCET values. Both implementations have been compiled in *release* mode (i.e. with all optimizations enabled), and run on the board at maximum clock speed (168 MHz).

We have observed some significant differences in execution times for certain operations between the Ada and Rust system. We ascribe these gaps to differences in I/O backends and runtime overheads,

since RTIC heavily relies on hardware features specific to ARM Cortex architecture, to compiler choices, and different math libraries.

NOTE: How do we explain the difference between rust and ada in execution whetstone? I think we should say that we used a very basic math library for Rust, only meant to be used as a fallback (libm) and not yet stable even. We used it to avoid including an external C library which would have complicated the compilation phase. While Ada uses a more optimized math library. I don't know if you have better ideas.

Operation	Rust (ns)	Ada (ns)
Small Whetstone (workload 139)	4346391	1018910
AL Read	136	1267
ALR Read	647	3256
Put Line	8017	8222

Table 3: WCET per operation — Activation Log Reader

4.3.1 Activation Log Reader.

Operation	Rust (ns)	Ada (ns)
AL Write	583	1678
EES Write	1071	3322

Table 4: WCET per operation — External Event Server

4.3.2 External Event Server.

Operation	Rust (ns)	Ada (ns)
RB Extract	303	1361
Extract Workload	814	4760
Small Whetstone (workload 278)	8692782	2037820
Put Line	7970	7406

Table 5: WCET per operation — On-Call Producer

4.3.3 On-Call Producer.

Operation	Rust (ns)	Ada (ns)
Small Whetstone (workload 756)	23639364	5541700
Due Activation	214	1344
RB Deposit	2398	3583
OCP Activation	2903	5283
Check Due	208	1372
ALR Signal	1982	6922
Cancel Deadline	273	2835
RP Cancel Deadline	8314	—

Table 6: WCET per operation — Regular Producer

4.3.4 Regular Producer.

Operation	Rust (ns)	Ada (ns)
Systick Overhead	702	3906
ISR Switch	2732	2850
Delay Until	865	5420
Signal RTIC	4160	–
Task Semaphore	4178	–
Event Queue	4101	–
Spawn Overhead	148	–
Context-Switch	166	3090

Table 7: WCET – Kernel/Runtime Overheads

4.3.5 Kernel/Runtime Overheads.

5 Deadline Miss Detection

In our system, each task is associated with a hard deadline, as described in Table [1].

In order to detect a deadline miss, since the *RTIC* framework does not provide any built-in mechanism for this, for each task we implemented a watchdog task responsible for monitoring job deadlines.

To achieve the desired behavior, i.e., a watchdog declares a deadline miss if and only if the associated job fails to complete its execution before its deadline, we used a shared resource, called `deadline_protected_object`. This resource is a struct containing a boolean flag, whose value represents whether the job has completed its execution or not.

In *RTIC* framework, each shared resource needs to be acquired before being accessed or before invoking any of its associated methods. Therefore, when a monitored job acquired the lock on the `deadline_protected_object` to invoke `cancel_deadline` (which is the method responsible for acknowledging the deadline), it can not be preempted by its watchdog task. This prevents the watchdog from preempting the monitored job while it is updating the flag, avoiding undesired deadline miss detections.

Additionally, the `deadline_protected_object` struct contains an activation counter, which allows for finer control over the requests to clear deadline misses, allowing stale requests to be ignored.

The watchdog tasks have the highest priority in the system: this ensures that as soon as a deadline is missed, the corresponding watchdog will be scheduled to run immediately, beside the aforementioned case in which the monitored job has already acquired the lock on `deadline_protected_object`. The highest priority also implies the `deadline_protected_object`'s ceiling is the highest value in the system, preventing the controlled task from blocking the watchdog's execution due to a higher ceiling value of a resource it may own.

Since in our system we have periodic and sporadic tasks, we implemented two different types of watchdogs:

- `periodic_deadline_watchdog`: this watchdog has as locals the period of the task and the next deadline of the monitored job. It delays itself until the next deadline and then, after

acquiring the shared `deadline_protected_object`, it invokes its method `deadline_miss_detected`. This method checks the flag and, if it is still false, it declares a deadline miss. Then, the watchdog increments the next deadline by the period of the associated task, and it loops again;

- `sporadic_deadline_watchdog`: this watchdog waits on an activation signal that notices it of a new activation of the monitored job. The watchdog then calculates the next deadline by adding the relative deadline of the associated task with the activation time of the monitored job, passed as argument inside the precedent signal. The watchdog then delays itself until the obtained absolute deadline and, after acquiring the shared `deadline_protected_object`, it invokes on it the method `deadline_miss_detected`, as happens in the `periodic_deadline_watchdog` case. After that, it loops again, waiting for the next activation signal.

To avoid the first activation of a periodic watchdog from starting before the monitored job can actually execute due to system initialization, we set its first deadline equal to the system activation instant plus the task's relative deadline.

The system activation instant is represented by a static `AtomicU32`, expressed in ticks, and serves as the reference point until which all tasks spawned during initialization are delayed. This mechanism provides the system with a well-defined "zero time".

Moreover, in order to provide accurate deadline management for sporadic tasks, their activation signals occur inside a critical section. This prevents unwanted preemptions between task activation and the sending of the signal to the corresponding watchdog, which could otherwise lead to an incorrect calculation of the next deadline.

6 MAST

The response time analysis of the system has been largely performed using the *MAST* tool [2]. *MAST* is a *Modeling and Analysis Suite for Real-Time* applications, developed at the *University of Canababria*. It facilitates the modelling of real-time systems and provides a set of tools for performing various types of analysis on a given model.

In particular, for the purpose of this project, *MAST* can perform response-time analysis for real-time systems on a single-processor using a *Fixed-Priority Scheduling* policy. The tool also supports preemptive tasks and the *Immediate Ceiling Protocol (ICP)* for managing shared resources. Since *ICP* and *SRP* exhibit similar observable behavior, *MAST* can be used to analyze *SRP*-based system, such as our *Rust* implementation.

6.1 MAST Model

To analyze the system using *MAST*, a model of the system must be provided in the form of an XML or text file. In this subsection, we describe the model components available for representing the system. Further details can be found in the *MAST description* [?].

6.1.1 Processing_Resource. It represents a resource capable of executing abstract activities. In our case, since we are dealing with a single-processor system, there is only one processing resource. When we define a processing resource, we must provide the range of priorities available, a *System_Timer* and the worst *ISR Switch* time. The latter parameter denotes the context switch time of an

interrupt service routine, without taking into account the execution time of the ISR itself.

6.1.2 System_Timer. It represents a timer used by a processing resource. In our system we have to deal with *Ticker*, a system that has a periodic ticker. The information required by this type of timer are the period and the worst-case overhead time, i.e., the time taken to process the ticker interrupt, including the scheduling overhead required for scanning and moving the jobs in the ready queue.

6.1.3 Scheduler. It represents the scheduling policy used to manage the amount of CPU processing capacity. As previously mentioned, our system uses a *Fixed-Priority Scheduling* policy. When defining a scheduler, we need to specify the processing resource it is hosted on, the available priority range and the worst context switch time. This last parameter denotes the time taken to set the context switch interrupt as pending and to execute its handler, which is responsible for saving registers of active context and for restoring the ones of the new active context.

6.1.4 Shared_Resource. It represents a resource that is shared among tasks. As stated before, the *MAST* model uses the *Immediate Ceiling Protocol*; therefore the priority specified for a shared resource is its ceiling priority, i.e. the highest priority of those tasks which access it.

6.1.5 Scheduling_Server. It represents a *schedulable* entity in a processing resource. Since we use a *Fixed-Priority Scheduling* policy, the server scheduling parameters are of type *Fixed_Priority_Policy*, which corresponds to regular preemptive fixed-priority scheduling, and requires a priority value to be specified.

6.1.6 Operation. It is an abstract representation of a piece of code. There are three types of operations:

- *Simple*: used to model simple operations or methods of protected objects. In the latter case, the operation requires a list of shared resources to lock before executing it and to unlock after its completion. It also requires the specification of best, average and worst-case execution times;
- *Composite*: an operation composed of an ordered sequence of other operations. Its execution times are derived from the execution times of its internal operations;
- *Enclosing*: similar to a composite one but we need to specify the best, average and worst-case execution times as well. This provides a way to express overheads associated with inner operations, such as the cost of acquiring and releasing a shared resource.

6.1.7 Transaction. It represents a modeling unit for concatenations of the system. Each transaction is composed of three components:

- *External_Events*: a list of events that models concatenations of the system with external devices through interrupts or with hardware timing devices. For each external event it is necessary to specify its type, such as periodic or sporadic, its period or minimum inter-arrival time, its jitter and its phase;
- *Internal_Events*: a list of events that are generated by event handlers. They can be used to express deadlines through *Timing_Requirements*: in our case, we use *Hard_Global_Deadline* timing requirements, which specify a hard deadline that

must be met by all instances of the internal event, and a reference to the external event that represents its activation;

- *Event_Handlers*: a list of event handlers that expresses relationships between events. Each event handler requires an event (or a list of events) in input and produces an event (or a list of events) in output. Based on the type of the event handler (such as *System_Timed_Activity*), it is also possible to execute an activity operation on a specified activity server.

6.2 MAST Tools

MAST supports different types of analysis tools. In this subsection, we present those used in our system's analysis, while further details can be found in the *MAST description* [3].

- *Classic RM Analysis*: the classic exact response-time analysis for single-processor fixed-priority systems. Although the name might suggest support only for *Rate Monotonic* scheme, it also covers *Deadline Monotonic* as well;
- *Offset-based Approximate with Precedence Relation Analysis*: response-time analysis that improves the holistic analysis by taking into account that tasks of the same transaction are not independent, through the use of offsets. Those precedence relations, together with tasks priorities, provide a tighter estimation of the response times comparing to other tools.

Each analysis tool, as described in the *MAST restrictions* [4], is subject to certain limitations regarding the expression power of the model, particularly in how transactions can be constructed. The most relevant are:

- *Restricted_Multipath_Transactions*: it imposes the absence of branch elements in side transactions;
- *Referenced_Events_Are_External_Only*: it requires only external events as referenceable events, therefore only external events can be used as activation events for deadlines;
- *Linear_Transactions_Only*: it allows just one external event per transaction.

7 FPS Analysis

In this section we present our *Fixed-Priority Scheduling (FPS)* analysis of the system.

The analysis was performed by modeling the system through a *MAST* model, which was then analyzed using the *MAST* tool to verify the timing properties and to analyze the system under different workloads.

Priority assignment was done by ensuring that tasks with shorter relative deadlines are given higher priorities, as in the *Deadline Monotonic* scheme, while also accounting for task relationships, since tasks in the system do not have implicit deadlines and presents some precedence relationships (further details can be found in the *Integration Report* [14]).

The evaluation of the system under different utilization levels was facilitated by the use of the *Small Whetstone* algorithm to simulate computational workloads. Thanks to its configurable load and deterministic behavior without involving the use of shared resources, it provides a straightforward way to test the system under different utilization levels by simply scaling the WCET in the different *small_whetstone* operations that are present in the *MAST* model, without the need for new execution-time measurements.

7.1 Holistic Analysis

We first conducted a holistic analysis, considering all the tasks as independent. As a consequence, this analysis is an over-approximation of the worst case, since it does not take into account any relationship between tasks, assuming a case that cannot occur in a real execution.

System feasibility is therefore evaluated based on the worst-case scenario at the critical instant, i.e., when all tasks are released simultaneously (not a realistic and possible case).

The analysis was performed using the *Classic RM Analysis* tool in *MAST*, which implements the classic exact response-time analysis for single-processor systems under *Fixed-Priority Scheduling* (although it's called *Rate Monotonic*, it bases *Deadline Monotonic* analysis).

The first results of the analysis were made considering the system under the loads provided in the original *Guided Extended Example*, i.e., with the following Whetstone workloads (accompanied by their corresponding WCETs expressed in seconds).

Task	Workload	Rust WCET	Ada WCET
Regular_Producer	756	0.023639364	0.00554170
On_Call_Producer	278	0.008692782	0.00203782
Activation_Log_Reader	139	0.004346391	0.00101891

Table 8: Whetstone Operations

The obtained results are reported in Table [9] and Table [13]. Times are expressed in seconds, and we omitted the watchdogs' transactions (which were instead included in the Rust model) since they are not relevant for identifying increasing workload factors. Regarding the jitter times, all transactions suffer jitter due to the system ticker interrupt and the context switch overhead. Moreover, additional sources of jitter and blocking times for each transaction are the following:

- *rp_transaction*: suffers additional jitter due to the possible interference of the *ees_transaction*. Its blocking time is instead due to the possible access to shared resources from the lower-priority *ocp_transaction* and *alr_transaction* with ceiling priority equal to the priority of *rp_transaction*;
- *ocp_transaction*: suffers additional jitter due to interference by the higher-priority *rp_transaction*. Its blocking time is caused by the *alr_transaction*;
- *alr_transaction*: suffers additional jitter due to interference by both the higher-priority *rp_transaction* and *ocp_transaction*. It does not suffer blocking, since it represents the lowest-priority task in the system;
- *ees_transaction*: it suffers no additional jitter than the aforementioned overheads. Its blocking time is caused by the *alr_transaction* which can possibly access the shared *Activation_Log* resource with ceiling priority equal to the priority of *ees_transaction*.

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.024728	2006.3%	0.001066	3.030E-06
ocp_transaction	0.033451	8791.0%	0.024741	2.730E-06
alr_transaction	0.037822	21927.3%	0.033455	0.000
ees_transaction	0.001043	>=100000.0%	2.955E-05	2.730E-06
System Utilization	2.79%			
System Slack	1994.1%			

Table 9: Holistic Analysis Results Rust

Based on these results, since the smallest slack available was 2006.3% for the *rp_transaction*, we increased the workloads by a factor of 20 to reduce it significantly. New results are reported in Table [10].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.474326	5.08%	0.001516	3.030E-06
ocp_transaction	0.648377	87.11%	0.474643	2.730E-06
alr_transaction	0.735413	303.91%	0.787757	0.000
ees_transaction	0.001043	>=100000.0%	0.001030	2.730E-06
System Utilization	53.76%			
System Slack	5.75%			

Table 10: Holistic Analysis Results Rust x20

We can notice how the blocking times have not changed, since protected operations remain the same. While we have achieved a small slack for the *rp_transaction* (5.08%), the other two transactions that involve Whetstone computations still have large available slacks. Based on these results, especially on the smallest slack of 87.11%, we decided to increase the *On_Call_Producer* and *Activation_Log_Reader* workloads by a factor 1.8, while keeping the *Regular_Producer* workload fixed. The updated results are reported in Table [11].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.474326	2.34%	0.001516	3.030E-06
ocp_transaction	0.787600	3.91%	0.473643	2.730E-06
alr_transaction	0.944248	35.55%	0.786757	0.000
ees_transaction	0.001043	91855.5%	0.001030	2.730E-06
System Utilization	58.86%			
System Slack	1.58%			

Table 11: Holistic Analysis Results Rust x1.8

Finally, based on the last useful available slack of 35.55% for the *alr_transaction*, we increased its workload by a factor of 1.35. The updated results are reported in Table [12].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.474326	0.00%	0.001516	3.030E-06
ocp_transaction	0.787600	0.00%	0.474643	2.730E-06
alr_transaction	0.999068	0.39%	0.787812	0.000
ees_transaction	0.001043	6911.7%	0.001030	2.730E-06
System Utilization	60.68%			
System Slack	0.39%			

Table 12: Holistic Analysis Results Rust x1.35

Even though the `ees_transaction` has still a large available slack, since it does not perform any Whetstone computation, it is not possible to increase the utilization of the system any further. Besides that, even a possible increase of the execution time of `ees_transaction` would not affect significantly the utilization of the system. In conclusion, under holistic analysis, we achieved 60.68% system utilization.

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.006624	8808.2%	0.001045	1.361E-06
ocp_transaction	0.007697	38006.3%	0.005638	1.267E-06
alr_transaction	0.008752	91394.9%	0.007706	0.000
ees_transaction	1.882E-05	$\geq 100000.0\%$	1.141E-05	1.267E-06
System Utilization	1.03%			
System Slack	7728.7%			

Table 13: Holistic Analysis Results Ada

Based on these results, since the smallest available slack was 8808.2% for the `rp_transaction`, we increased the workloads by a factor of 90 to reduce it significantly. We chose a factor of 90 instead of 88 by the fact that the *Small Whetstone* operation represent only a part of the transaction rather than the entire `rp_transaction`. The updated results are reported in Table [14].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.490670	1.56%	0.002963	1.361E-06
ocp_transaction	0.669740	72.27%	0.490390	1.267E-06
alr_transaction	0.759792	265.23%	0.670102	0.000
ees_transaction	1.882E-05	$\geq 100000.0\%$	1.141E-05	1.267E-06
System Utilization	55.74%			
System Slack	1.98%			

Table 14: Holistic Analysis Results Ada x88

Again, we notice how the blocking times have not changed, since protected operations remain the same. While we have achieved a small slack for the `rp_transaction` (1.56%), the other two transactions that involve Whetstone computations still have large available slacks. Based on these results, especially on the smallest slack of 72.27%, we decided to increase the `On_Call_Producer` and `Activation_Log_Reader` workloads by a factor of 1.7, while keeping the `Regular_Producer` workload fixed. The updated results are reported in Table [15].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.490670	0.78%	0.002963	1.361E-06
ocp_transaction	0.795769	1.17%	0.490889	0.000
alr_transaction	0.948836	32.42%	0.796380	1.361E-06
ees_transaction	1.882E-05	56921.9%	1.141E-05	1.267E-06
System Utilization	60.35%			
System Slack	0.78%			

Table 15: Holistic Analysis Results Ada x1.7

Finally, based on the last useful slack available of 32.34% for the `alr_transaction`, we increased its workload by a factor of 1.3. The updated results are reported in Table [16].

Transaction	R_{max}	Slack	Jitter	MBT
rp_transaction	0.490670	0.78%	0.002963	1.361E-06
ocp_transaction	0.795769	1.17%	0.490889	0.000
alr_transaction	0.994747	1.95%	0.796563	1.361E-06
ees_transaction	1.882E-05	56921.9%	1.141E-05	1.267E-06
System Utilization	61.87%			
System Slack	0.78%			

Table 16: Holistic Analysis Results Ada x1.3

Again, the considerations made before upon the `ees_transaction` apply here. In conclusion, under holistic analysis, we achieved 61.87% system utilization.

7.2 Offset-Based Analysis

To improve the results obtained with the holistic analysis, we performed an offset-based analysis, which takes into account the relationships between tasks, thus providing less pessimistic results.

To exploit the benefits of offset-based analysis, we use the *Offset-Based Analysis with Precedence Relation* tool provided in *MAST*. Due to restrictions imposed by this model, we had to represent each periodic `regular_producer` activation as a separate transaction, plus the additional transaction required for the `external_event_server`. The `on_call_producer` and `activation_log_reader` activations are included within the `rp_transactions` when needed, exploiting the system-imposed order, i.e., the `activation_log_reader` always executes after the `on_call_producer`, which, in turn, always executes after the `regular_producer` (if they are activated in the same period). Since the hyperperiod of the system is 15 seconds, we have 15 `rp_transactions` plus the `ees_transaction`, resulting in a total of 16 transactions.

If the offset-based analysis helps reduce the interference between tasks, we are unable to increase the workloads beyond what was achieved with the holistic analysis. The reason is that the critical instant faced in the holistic analysis is similar to the one faced with the offset-based analysis in the 12th period, when the `regular_producer` activates both the `on_call_producer` and the `activation_log_reader`. A graphical representation of this scenario is shown in Figure [2], where we omitted the inclusion of `ees_transaction`.

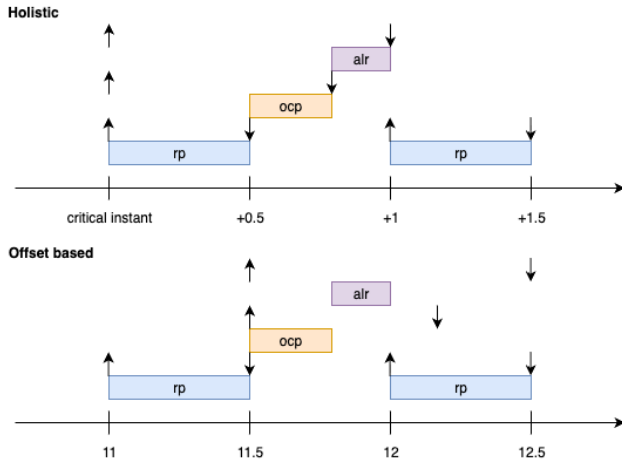


Figure 2: Holistic Analysis vs Offset-Based Analysis at the 12-th Period

The figure shows how the workloads obtained with the holistic analysis already fill the available slacks also in the offset-based analysis.

To further increase the utilization of the system, we decided to take advantage of those tasks which activate more frequently during the hyperperiod. While the regular_producer, which activates every second, already executes for almost all of its executable period time, until its deadline, we noticed how the activation_log_reader, which has a period of 3 seconds, has a workload that is lower than the one of the regular_producer, even though it has a period of 5 seconds. This implies that, during the hyperperiod, the activation_log_reader activates 5 times, whereas the on_call_producer activates only 3 times. Of these activations, only in the 12th period do both tasks execute in the same period. Moreover, since the regular_producer has a period of one second and triggers both the on_call_producer and activation_log_reader, and considering that it has a higher priority than both, the available execution time for these two tasks is limited to 0.5 seconds, even though their respective deadlines (0.8 and 1 second) would allow for longer executions.

Since our goal is to maximize the total utilization, we need to increase as much as possible to fill the identified 3.5 seconds (there are 8 intervals due to 8 total activations, but since one is shared, we obtain 7 intervals of 0.5 seconds each).

Before expressing the situation as an optimization problem, several considerations must be taken into account. First, we need to account for the system ticker overhead, which can be estimated as:

Tot Sys Ticker Overhead = $\frac{0.500}{0.001} * \text{sys ticker worst overhead}$
Next, we must consider the dispatching cost that occurs when the on_call_producer completes and the activation_log_reader begins its execution. This overhead is pessimistically estimated as the worst-case context switch time.

Finally, we need to consider the possible execution of the ees_transaction: since the external_event_server has an higher priority than both the tasks (this is not true for the

activation_log_reader if it had acquire the activation_log, thanks to the Stack Resource Protocol), we must account for the cost of preemption (i.e., the context switch) plus the WCET of ees_transaction.

We end up with the following available times for the two systems:

- Rust System: $0.5 - 0.0005 - 0.018\text{E-}06 - (0.018\text{E-}06 + 13.486\text{E-}06) = 0.49948648$ seconds;
- Ada System: $0.5 - 0.001928 - 3.964\text{E-}06 - (3.964\text{E-}06 + 4.303\text{E-}06) = 0.49805977$ seconds.

The problem can now be expressed as the following linear optimization problem:

Let x = execution time of on_call_producer

y = execution time of activation_log_reader

T = available time: 0.49948648 s (Rust), 0.49805977 s (Ada)

Maximize $3x + 5y$

Subject to $x \leq T$

$y \leq T$

$x + y \leq T$

$x \geq \text{WCET on_call_producer with workload 1}$

$y \geq \text{WCET activation_log_reader with workload 1}$

For both systems, the solution to this linear optimization is $x = \text{on_call_producer with workload 1}$, with y that fills the remaining available time slot in T , resulting in $0.49948648 - 0.00004456 = 0.49944192$ seconds for the *Rust* system and $0.49805977 - 0.00002079 = 0.49785182$ seconds for the *Ada* system. Here, 0.00004456 and 0.00002079 correspond to the WCETs of the on_call_producer in *Rust* and *Ada*, respectively, which include not only the execution of a single Whetstone operation but also all the other operations that the job must normally perform.

With these updated workloads, the offset-based model, with the *Offset-Based Analysis with Precedence Relation* tool in *MAST*, yields the following results:

- Rust Total Utilization: 64.03%;
- Ada Total Utilization: 65.77%.

Since the transactions have the same activation source (a regular_producer job), there is no interference between tasks. Therefore, to validate our assumptions, we just consider only whether best-case response times are within the deadlines, ignoring the worst-case response times as well as slacks.

8 Experimental Results

To prove the validity of our response-time analysis, we ran some experiments on the software application on a real development board (*Olimex STM32H405*). In particular, the goal was to validate the scheduling feasibility of the system using the computed workloads from the RTA, and then obtain the values that make the system fail to meet its deadlines (*sensitivity analysis*). The latter experiments were only performed on the Rust system.

8.0.1 Holistic Analysis. From the holistic analysis, we obtained the workloads reported in Table [17] that maximize the system utilization. With this computed workloads both systems managed

Task	Workload (ops)	Scaling
Regular_Producer	15120	756×20
On_Call_Producer	10008	$278 \times 20 \times 1.8$
Activation_Log_Reader	6755	$139 \times 20 \times 1.8 \times 1.35$

Table 17: Holistic baseline workloads (board experiments)

Scenario	RP (ops)	OCP (ops)	ALR (ops)	Observed
Baseline	15120	10008	6755	No deadline misses
RP increase	15200	10008	6755	ALR misses $\sim 1/4$ (when released with OCP)
RP increase	16000	10008	6755	RP always misses; ALR misses as before
OCP increase	15120	10100	6755	ALR misses $\sim 1/4$ (when released with OCP)
OCP increase	15120	16900	6755	OCP always misses; ALR misses as before
ALR increase	15120	10008	6900	ALR misses $\sim 1/4$
ALR increase	15120	10008	16900	ALR always misses; no impact on others

Table 18: Holistic sensitivity results (board experiments)

Task	Workload (ops)	Notes
Regular_Producer	15120	Baseline from analysis
On_Call_Producer	15973	Baseline from analysis
Activation_Log_Reader	1	Baseline from analysis

Table 19: Offset-based baseline workloads (board experiments)

Scenario	RP (ops)	OCP (ops)	ALR (ops)	Observed
Baseline	15120	1	15973	No deadline misses
RP increase	16000	1	15973	RP and ALR always miss
OCP increase	15120	900	15973	ALR misses $\sim 1/4$ (when released with OCP)
OCP increase	15120	16900	15973	OCP always misses; ALR misses as before
ALR increase	15120	15973	16900	ALR always misses; no impact on others

Table 20: Offset-based sensitivity results (board experiments)

to meet all the deadlines after execution of 50 *hyperperiods*, which was considered to be a sufficient number of tests.

Then, we started experimenting with varying workloads by focusing on one task at a time keeping the others fixed, we obtained the results in Table [18]. Workloads values were incremented at steps of 100.

Results matched closely the static analysis performed, since even small increments from the original baseline incurred in deadline misses. These results also highlight a key property of FPS: under overload conditions the tasks gets affected in priority order (from lower to highest).

8.1 Offset-based Analysis

Using offset-based analysis we obtained the workloads in Table [19] that maximize the system utilization.

No deadline misses were observed after 10 *hyperperiods*.

The sensitivity analysis produced the values in Table [20].

This time, results were slightly more distant from the ones obtained from static analysis. This could be due to an overestimation of the WCET of system and operation overhead.

Nonetheless, results still follow a consistent behavior when overload is reached, with the lowest priority jobs suffering from interference and incurring in deadline misses.

9 Conclusions

In this technical report, we presented the implementation of a real-time system in Rust, and we compared it with an equivalent implementation in Ada. The analysis, conducted with the *MAST* tool, showed that both implementations are able to achieve similar results in terms of system utilization. To support the analysis, both systems were also tested on an *Olimex STM32H405* [20] using the identified workloads, and no missed deadlines were observed. Then we tested the validity of our analysis, in terms of "pessimism" (sensitivity), obtaining consistent results.

Rust, and in particular the *RTIC* framework, proved to be a valid alternative for the development of real-time systems. The *RTIC* framework, being a declarative system model for single-core applications, enforces a well-defined structure, requiring a clear understanding and strict control over the management of both local and shared resources. This strong structure can support developers in maintaining a well-organized and systematic design of the system.

However, in some situations it may also limit design flexibility in relation to *Ada Ravenscar Profile*, especially when dealing with shared resources. In particular, we were unable to separate the responsibility of the `on_call_producer` activation, as it was not possible to implement it except by embedding it directly within the `deposit` call on the `Request_Buffer`. This limitation stems from the fact that any access to a shared resource must be explicitly declared in the task's shared resources and, consequently, it is not possible for a task to provide an interface through which other tasks may interact with. The management of shared resources in *RTIC* always requires acquiring a lock when accessing them: even if *Stack Resource Policy* (SRP) is lock-free, the lock acquisition is used to delimit critical sections in which tasks can access the acquired shared resources and invoke on them their methods, so that the runtime can perform the required analysis as explained in section 3.1. In contrast, *Ada Ravenscar Profile* achieves the same goal through protected objects, which automatically manage resource access without requiring explicit lock acquisition to delimit critical sections. Once again, this enforced structure represents the trade-off between flexibility and the strong safety guarantees provided by *Rust*. Moreover, the lack of guard mechanisms for shared resources in *RTIC* may lead to potential runtime errors if not properly managed by developers. This requires designing the system carefully, with resources that may need to notify interested tasks upon status changes. In such cases, the responsibility is shifted to developers, forcing resources to be aware of the tasks that depend on them and thereby increasing the coupling between resources and tasks. By contrast, *Ada Ravenscar Profile* inherently prevents such situations through its *entry* mechanism, providing more suitable abstractions for managing shared resources.

Unlike the *Ada Ravenscar Profile*, on the subject of deadline miss detections, the *RTIC* framework reveals a significant limitation by providing no built-in support for handling such events, thus it leaves the responsibility for handling timing violations entirely to the developers.

Overall, we consider the *RTIC* framework to be a valid, flexible and performant option for the development of real-time systems, with a generated code that, as shown in this report, infers no substantial overhead compared to an equivalent implementation in *Ada Ravenscar Profile*.

The code relevant for this work is available on GitHub repositories, the URLs can be found in the references section. The *rtks* repo [22] contains the *Rust* system code, while the *gee* repo [21] contains the *Ada* one. The identification of some *RTIC* system overheads was performed in the *Profiling-RTIC* repository [13].

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